

Maths — Ratio Retest

12 questions · Sections 1–3

Name: _____

Date: _____

For every question:

1. Show your working.
2. Write the unit (\$, mL, L, ₹).

Section 1

Q1.1

A 1 · C 1

Write the ratio in its lowest terms.

63 : 49

Q1.2

A 2 · C 1

Simplify the ratio to its lowest or simplest form.

$2 : \frac{2}{7}$

Q1.3

A 2 · C 1

Three friends Edrian, Ethan and Kurt go shopping.

The money they each have is in the ratio **Edrian : Ethan : Kurt = 5 : 7 : 8**.

Edrian has **\$15**.

How many dollars do they have in total?

Q1.4

A 2 · C 1

Three friends Edrian, Ethan and Kurt go shopping.

The money they each have is in the ratio **Edrian : Ethan : Kurt = 5 : 7 : 8**.

If Edrian has **\$15**, how many dollars do Ethan and Kurt have?

Q1.5

A 2

Three friends Edrian, Ethan and Kurt go shopping.

The money they each have is in the ratio **Edrian : Ethan : Kurt = 5 : 7 : 8**.

If Edrian had **1 dollar**, Ethan and Kurt would have how much?

Section 2**Q2.1**

A 1

Round **5363.89** to **4 significant figures**. Circle one.

A 5300.00 B 6.400 C 6400 D 5364 E 5.400

Q2.2

A 2

Sort these numbers in order of size, starting with the smallest.

0.8 83% 0.72 $\frac{3}{4}$

Q2.3

A 2 · C 2

A recipe says to mix lemon juice to lime juice in the ratio **5 : 2**.

- a) If a **70 mL** mixture is made, calculate how much is lime juice.
b) If **75 mL** of lemon juice is used, find out how much lime juice should be added.

Q2.4

A 6 · C 6

Ben and Alice share the cost of a laptop computer in the ratio **5 : 8**.

- a) If the laptop costs **\$2600**, how much will each person pay?
b) If Ben pays **\$450**, how much will Alice pay?
c) If Alice pays **\$600**, how much does the laptop cost in total?

Q2.5

A 2 · C 2

Leon has two buckets of lemons and limes, each containing **40** pieces in total. The first bucket contains **31 lemons**. If the overall ratio of lemons to limes is **5 : 3**, how many limes are in the second bucket?

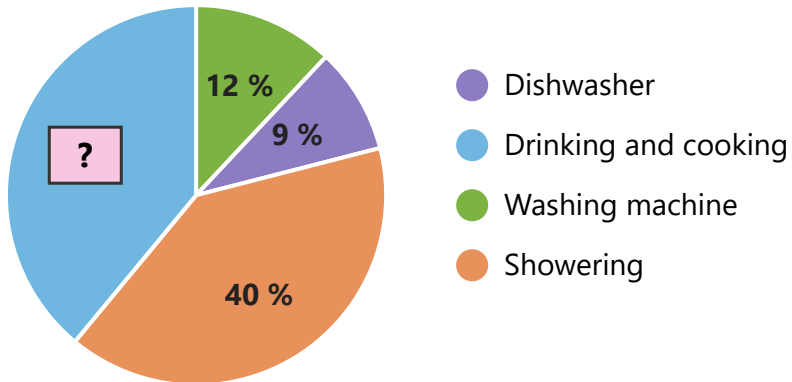
Section 3

Q3.1

D 2

Write down the missing **percentage** for drinking and cooking on the pie chart.

Daily percentage of water usage per person



Q3.2

C 8 · D 8

Planning a School Fruit Drink

The student council is preparing a fruit drink for a school event. The recipe uses **orange juice and apple juice in the ratio 3 : 2**. They have **18 litres of orange juice** available.

(a) Determine how many litres of apple juice are needed to maintain the correct ratio. [2 marks]

(b) Determine the amount of orange juice and apple juice needed to make exactly **50 litres** of fruit drink while maintaining the ratio 3 : 2. Show your working. [3 marks]

The school can buy apple juice in two different containers:

Option A: 2 litres for ₹180

Option B: 5 litres for ₹425

(c) Using your answer from question (b), determine which option would be the **better choice** for the school if they want to buy enough apple juice while spending as little as possible. Explain your reasoning. [3 marks]

A student says: *"Option B is always better because you get more juice for your money."*

(d) Do you agree with the student's statement? Use **mathematical evidence** involving ratios or unit rates to justify your answer. [2 marks]